

LOVELY PROFESSIONAL UNIVERSITY

Academic Task No. 1

School: SCA

Faculty of: LFTS

Course Title: PROGRAMMING IN JAVA-LAB

Course Code:CAP680

Max. Marks: 30[10 Marks each]

Date of Allotment: 14th Feb. 2022

Last Date of Submission: 23rd Feb. 2022

Important Guidelines:

1. All questions in this Academic Task are compulsory.
 2. It is mandatory to attempt all questions of the assignment in your own handwriting on A4 size sheets/pages with a blue color ink pen. Any other mode of attempt (typed or printed codes or table) except hand written/drawn will not be accepted/considered as valid submission(s) under any circumstances.
 3. Every attempted sheet/page should carry clear details of student such as Name, Registration number, Roll number, Question number and Page number. The page numbers should be written clearly on the bottom of every attempted sheet in a prescribed format as: for page 1; **Page 1 of 4**, for page 2; **Page 2 of 4**, for page 3; **Page 3 of 4** and for page 4; **Page 4 of 4**, in case your assignment/document is of 4 pages.
 4. After attempting the answer(s), student needs to take photograph of each of these answer sheets/pages and needs to convert the **jpeg** format images into a sequential single **pdf** format document (can be done with many free online available converters).
 5. This PDF file should be uploaded onto the UMS interface on or before the last date of the submission.
 6. Refrain from indulging into plagiarism as copy cases will be marked zero
 7. For every program it is mandatory to give proper documentation.
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SET-1

Q1: Given that an EMPLOYEE class contains the following members: **Date Members:** EmployeeID, EmployeeName, BasicSalary, DA, IncomeTax, NetSalary **Member Functions:** to read data, to calculate NetSalary and to print data members; Write a program to read data of 5 employees and compute the NetSalary of each employee (DA = 52% of Basic and Income Tax = 30% of the gross salary).

Q2: Write a program of sorting an array. Declare single dimensional array and accept 5 decimal values from the user. Then sort the input in descending order and display output.

Q3: Write a Java program that prints all real solutions to the quadratic equation $ax^2+bx+c = 0$. Read in a, b, c and use the quadratic formula. If the discriminate b^2-4ac is negative, display a message stating that there are no real solutions?

SET-2

Q1: Write a program to perform following operations on a given number and check whether result value is prime or not?

- a. Sum of given number
- b. Product of given number

For e.g: Given number is: 1231

Sum of number is:(1+2+3+1) 7 is a prime no.

Product of number is:(1*2*3*1) 6 is not a prime no.

Q2: Write a program to display a color name depending on color value using switch case. Ask color value from user.

For e.g: Color Value: #1111

Color Name: White

Color Value: #0000

Color Name: Black

Q3: Write a program to create two multidimensional arrays of same size. Accept value from user and store them in first array. Now copy all the elements of first array are second array and print output.

SET-3

Q1: Suppose there is the string "Lovely Professional University" define some methods which will be use to perform

- a. finds the total no. of characters
- b. finds the position of string University
- c. extracts the string Lovely
- d. replaces Lovely with lovely (first letter 'l' in small case)
- e. changes all characters in the string in Upper case.

Q2: Create a class named 'Time' with data members (hours, mins, secs). Define methods in this class named 'hourConvertor', 'minConvertor' and 'secConvertor'. Calculate total hours, mins and secs for any integer value (in days).

For eg:

Input : 2 (days)

output: total_hours:48

total_mins:2880

total_secs:172800

Q3: Write a Java program that checks whether a given string is a palindrome or not.

Ex: MADAM is a palindrome?

SET-4

Q1: Declare an array of 5 STUDENT objects. Using appropriate functions, find the average of the two better marks for each student.

Q2: The Fast Freight Shipping Company charges the following rates:

Weight of Package (in Kilograms) Rate per 500 Miles shipped

2 KG or less: \$1.10

Over 2 Kg but not more than 6 Kg: \$2.20

Over 6 KG but not more than 10 Kg: \$3.70

Over 10 KG but not more than 20 KG: \$4.80

Write a program that asks for the weight of the package and the distance it is to be shipped, and then displays the charges.

Input validation:

- Do not accept values of 0 or less for the weight of the package.
- Do not accept weights of more than 20 Kg (this is the maximum weight the company will ship).
- Do not accept distances of less than 10 miles or more than 3,000 miles.

Q3: Write a Java program to multiply two given matrices?

SET-5

Q1: Suppose you are planning to go to park so you are going to check tickets criteria online. The ticket rates details have been given

*If children below 10 are not allowed to swing

*If age is between 10 to 15 allowed to swing and getting 10 % discount

*If age is between 15 to 20 allowed to swing and getting 5 % discount

*If age is more than 20 not then not eligible for swing and discount

The age of person will run until you enter the age of last family member and then calculate the total charge amount after entering each person's age. Assume price of ticket is 100 Rs. each person.

Q2: Write a program in Java, define a STUDENT class with RegNo, Name and Marks in 3 CA of a subject. Define one function to find the best two CA marks for student.

Q3: You have to generate electricity bill for a customer with the details as give below:

Customer ID, Customer Name, Customer Address, Contact No, Units Consumed, Total payable amount. Following are the norms to calculate the unit amount:

- a) for initial 100 units it will cost 6 rupees per unit
- b) for next 100 units it will cost 8 rupees per unit
- c) for next all units it will cost 10 rupees per unit

SET-6

Q1: Write a program that computes the tax and tip on a restaurant bill for a patron with a \$44.50 meal charge. The tax should be 6.75% of the meal cost. The tip should be 15% of the total after adding the tax. Display the meal cost, tax amount, tip amount, and total bill on the screen.

Q2: A Booksellers has a book club that awards points to its customers based on the number of books purchased each month. The points are awarded as follows:

- If a customer purchases 0 books, 0 points are earned.
- If a customer purchases 1 book, 5 points are earned.
- If a customer purchases 2 books, 15 points are earned.
- If a customer purchases 3 books, 30 points are earned.
- If a customer purchases 4 or more books, 60 points are earned.

Write a program that asks the user to enter the number of books that he or she has purchased this month and then display the number of points awarded.

Q3: Swati bought 600 shares of stock at a price of \$21.77 per share. She must pay her stock broker a 2% commission for the transaction. Write a program that the display the following:

- The amount paid for the stock alone (without the commission)
- The amount of the commission
- The total amount paid

SET-7

Q1: Create an app in which user enter his NTS and F.Sc marks and your program will help student in selection of university. Based on these marks Student will be allocated a seat at different department of different university

University Name	Departments	Criteria
Oxford University	IT	Above 70% in Fsc. and 70 % in NTS
Oxford University	Engg	Above 70% in Fsc. and 60 % in NTS
Oxford University	Telecommunication	Above 70% in Fsc. and 50 % in NTS
MIT	IT	70% -60 % in Fsc. and 50 % in NTS
MIT	Engg	59% – 50 % in Fsc. and 50 % in NTS
MIT	Computer	Above 40% and below 50 % in Fsc. and NTS

Q2: Create a class named 'Time' with data members (hours, mins, secs). Define methods in this class named 'hourConvertor', 'minConvertor' and 'secConvertor'. Calculate total hours, mins and secs for any integer value (in days).

For eg:

Input : 2 (days)

output: total_hours:48

total_mins:2880

total_secs:172800

Q3: Suppose there is XYZ Company and there are different departments like production, marketing, finance, sales etc. Manager of the company want to know about the detail of the employees who are highly paid. Write a program using the concept of classes to implement the same.

SET-8

Q1: Write a Java program that prints all real solutions to the quadratic equation $ax^2+bx+c = 0$. Read in a, b, c and use the quadratic formula. If the discriminate b^2-4ac is negative, display a message stating that there are no real solutions?

Q2: Write a program to display a color name depending on color value using switch case. Ask color value from user.

For e.g: Color Value: #1111

Color Name: White

Color Value: #0000

Color Name: Black

Q3: Declare an array of 10 STUDENT objects. Using appropriate functions, find the average of the two better marks for each student.

SET-9

Q1: The Fast Freight Shipping Company charges the following rates:

Weight of Package (in Kilograms) Rate per 500 Miles shipped

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Over 6 KG but not more than 10 Kg: \$3.70

Over 10 KG but not more than 20 KG: \$4.80

Write a program that asks for the weight of the package and the distance it is to be shipped, and then displays the charges.

Input validation:

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-The amount paid for the stock alone (without the commission)

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-The total amount paid

Q3: Suppose there is the string "Lovely Professional University" define some methods which will be use to perform

- finds the total no. of characters
- finds the position of string University
- extracts the string Lovely
- replaces Lovely with lovely (first letter 'l' in small case)
- changes all characters in the string in Upper case.

List of Students:

Sr. No.	Reg. Number	Name	RollNumber	SET
1	12111790	Shivam Kumar Choudhury	RD2110A02	Set-1
2	12111810	Satyam	RD2110A03	Set-2
3	12112542	Kanav Jasrotia	RD2110A04	Set-3
4	12112743	Himanshi Gupta	RD2110A05	Set-4
5	12113075	Ankit Kumar Mishra	RD2110A07	Set-5
6	12113094	ANKIT KUMAR SONI	RD2110A08	Set-6
7	12110164	Vishal Ojha	RD2110A10	Set-7
8	12110187	Renbithung M Ngullie	RD2110A11	Set-8
9	12110286	Himansu Kumar Das	RD2110A12	Set-9
10	12110287	Manish Kumar Tripathy	RD2110A13	Set-1
11	12110266	Milan Ray	RD2110A14	Set-2
12	12110310	Nitish Sharma	RD2110A15	Set-3
13	12110150	Hetai Sood	RD2110A16	Set-4
14	12106059	Sachin Pal	RD2110A17	Set-5
15	12108431	Kajal Kumari	RD2110A18	Set-6
16	12105453	Manish Kumar Sharma	RD2110A19	Set-7
17	12105973	Harsh Sood	RD2110A21	Set-8
18	12105965	Dibyanshu Kumar Patra	RD2110A22	Set-9
19	12105383	Ananya Guha	RD2110A23	Set-1
20	12105389	Prabhat Kumar	RD2110A24	Set-2
21	12105410	Mwelwa Bangwe	RD2110A25	Set-3
22	12101308	Sandeep Kumar Gupta	RD2110A26	Set-4
23	12101712	Mohammad Abdul Samad	RD2110A27	Set-5
24	12101723	Anshu kumar	RD2110A28	Set-6
25	12101696	Vikas Pal	RD2110A29	Set-7
26	12101673	Arjit Biswal	RD2110A30	Set-8
27	12101666	Monika Rastogi	RD2110A31	Set-9
28	12101626	Archana Kumari	RD2110A32	Set-1
29	12101663	Ashish Kaushal	RD2110A33	Set-2
30	12101057	Abinash Prusty	RD2110A34	Set-3
31	12101059	Kanhugopal Dutta	RD2110A35	Set-4

32	12101060	Subhankar Mallick	RD2110A36	Set-5
33	12100685	Kiran Panwar	RD2110A37	Set-6
34	12100174	Anchal Sagar	RD2110A38	Set-7
35	12100232	Satish Kumar Choudhary	RD2110A39	Set-8
36	12100272	Gulam Nabi	RD2110A40	Set-9
37	12100268	Avinash Kumar	RD2110A41	Set-1
38	12100165	Dolly Kumari	RD2110A42	Set-2
39	12100139	Manish Das	RD2110A43	Set-3
40	12100375	Radha Gobinda Das	RD2110A44	Set-4
41	12100417	Neha Singh	RD2110A45	Set-5
42	12100353	Kamalkant	RD2110A46	Set-6
43	12100524	Shiv Kumar Singh Gautam	RD2110A47	Set-7
44	12100442	Ajay Kumar Upadhyay	RD2110A48	Set-8
45	12100459	Shahid Sadiq	RD2110A49	Set-9
46	11611147	Anindita Chakraborty	RD2110A99	Set-1
47	12100180	Muddam Balaji	RD2110B50	Set-2
48	12100837	Shivam Shukla	RD2110B51	Set-3
49	12102710	Abhay Khajuria	RD2110B52	Set-4
50	12102715	Rahul Syal	RD2110B53	Set-5
51	12102635	Harpreet Kaur	RD2110B54	Set-6
52	12102541	Sonia	RD2110B55	Set-7
53	12102530	Aryan Mot	RD2110B56	Set-8
54	12102924	Ajay Upadhyay	RD2110B57	Set-9
55	12102983	Atul Kumar Gupta	RD2110B58	Set-1
56	12102947	Ishika Keshwani	RD2110B59	Set-2
57	12103005	Raju Basak	RD2110B60	Set-3
58	12102850	Abhinav Kumar	RD2110B62	Set-4
59	12102866	Sheetal	RD2110B63	Set-5
60	12102875	Trina Mahapatra	RD2110B64	Set-6
61	12102895	Nishant Raj Singh	RD2110B65	Set-7
62	12102899	Nitin Shukla	RD2110B66	Set-8
63	12102200	Abhinay Sharma	RD2110B67	Set-9
64	12102231	Deepak Kumar	RD2110B68	Set-1
65	12102235	Nitesh Khatri	RD2110B69	Set-2
66	12102245	Ashwani Kumar	RD2110B70	Set-3
67	12102284	Rohan Sharma	RD2110B71	Set-4
68	12102282	Thota Venkata Sai	RD2110B72	Set-5
69	12102289	Vivek Kumar	RD2110B73	Set-6
70	12102293	Aadil Ahmad Yatoo	RD2110B74	Set-7
71	12102130	Amit Babani	RD2110B75	Set-8
72	12102134	Akhil Kumar Verma	RD2110B76	Set-9

73	12102137	Saksham Verma	RD2110B77	Set-1
74	12102011	Rishu Raj	RD2110B78	Set-2
75	12101985	Shubham Kumar Singh	RD2110B79	Set-3
76	12101974	Shivam Kumar	RD2110B80	Set-4
77	12102006	Praveen Gaur	RD2110B81	Set-5
78	12102076	Divya Bharati	RD2110B82	Set-6
79	12101922	Rupali Verma	RD2110B83	Set-7
80	12101865	Saurabh Kumar Jha	RD2110B84	Set-8
81	12105482	Shubham Roy	RD2110B85	Set-9
82	12103820	Sumit Soni	RD2110B86	Set-1
83	12109446	Md Sagir	RD2110B87	Set-2
84	12109288	Navedeet Kumar	RD2110B88	Set-3
85	12109105	Rakesh Kumar Sharma	RD2110B89	Set-4
86	12108949	KM Geetika Pandey	RD2110B90	Set-5
87	12108912	Aakash Kalra	RD2110B91	Set-6
88	12110238	Krishna Sharma	RD2110B92	Set-7
89	12111090	Yuvraj Singh	RD2110B93	Set-8
90	12110972	Gaurav Kumar	RD2110B94	Set-9
91	12111026	Rishav Sagar	RD2110B95	Set-1
92	12112360	Rishu Kumari	RD2110B96	Set-2
93	12111504	Raviranjana Kumar	RD2110B97	Set-3